

# Electricity Demand Peak-Valley Analysis Using MySQL with Storage Use Case Implications

*An Independent SQL Portfolio Project Based on CAISO Hourly Electricity Demand Data*

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Tools: MySQL, Python, Excel

Data Source: EIA Hourly Electric Grid Monitor

Region: CAISO

Period: July 2024

## 1. Project Background

Electricity demand is not evenly distributed throughout the day. In most power systems, demand tends to be lower during off-peak hours and higher during peak-load periods. This intraday fluctuation creates operational challenges for grid operators, because electricity supply and demand must be balanced in real time. When the gap between daily peak demand and valley demand becomes large, the system may require more flexible resources to maintain reliability and reduce pressure during peak hours.

Battery energy storage is one potential solution to this problem. Storage systems can charge during low-demand periods and discharge during high-demand periods, helping to smooth demand fluctuations and support peak shaving and valley filling. Therefore, analyzing hourly electricity demand patterns is a useful first step for understanding where storage may create value.

This project uses hourly electricity demand data from the California Independent System Operator (CAISO) in July 2024. By using MySQL to analyze peak-valley gaps, hourly demand patterns, forecast errors, and weekday-weekend differences, the project aims to identify key demand characteristics and connect them with potential storage use cases.

## 2. Data Source and Processing

The dataset used in this project was obtained from the U.S. Energy Information Administration (EIA) Hourly Electric Grid Monitor. The selected balancing authority is the California Independent System Operator (CAISO), and the analysis period is July 2024. The raw dataset contains hourly electricity operation records, including actual demand, day-ahead demand forecast, net generation, and total interchange.

The raw data was downloaded through the EIA API and saved as a CSV file. Before importing the data into MySQL, a simple Python script was used to clean the column names. For example, columns such as respondent-name, type-name, and value-units were renamed as respondent\_name, type\_name, and value\_units to make them easier to query in SQL.

After cleaning, the CSV file was imported into MySQL Workbench. The raw table contains 2,980 records, with four major data categories: actual demand, day-ahead demand forecast, net generation, and total interchange. Since the main objective of this project is to analyze demand-side peak-valley characteristics, a separate SQL view was created to keep only actual demand records for July 2024. This demand view contains 744 hourly observations, corresponding to 31 days multiplied by 24 hours.

**Table 1. Dataset Overview**

| Item                | Description                                    |
|---------------------|------------------------------------------------|
| Data source         | EIA Hourly Electric Grid Monitor               |
| Region              | California Independent System Operator (CAISO) |
| Period              | July 2024                                      |
| Frequency           | Hourly                                         |
| Raw records         | 2,980                                          |
| Demand View Records | 744                                            |
| Tools               | Python, MySQL Workbench, Excel                 |

### 3. SQL Methodology

The analysis was mainly conducted in MySQL Workbench. After importing the cleaned CSV file into MySQL, the first step was to check the data structure, total row count, and major data categories. The raw table included four types of hourly records: actual demand, day-ahead demand forecast, net generation, and total interchange.

To focus on demand-side analysis, a SQL view named caiso\_demand\_july2024 was created. This view filtered the raw table to keep only actual demand records, converted the original time field into a usable datetime format, and extracted the date and hour information for later analysis. This step made the following queries easier to write and reduced repeated filtering conditions.

Several SQL techniques were used in this project. GROUP BY was used to calculate daily peak demand, valley demand, peak-valley gaps, and average hourly demand. CASE WHEN was used to classify records by weekday and weekend. Window functions, especially RANK() OVER(), were used to identify the exact peak and valley hour for each day. In addition, the actual demand and day-ahead forecast records were combined to calculate forecast errors and absolute forecast errors.

Overall, the SQL workflow was designed to move from raw hourly data to interpretable demand indicators. The main outputs include daily peak-valley gaps, hourly demand patterns, peak and valley hour distributions, forecast error summaries, and weekday-weekend demand comparisons.

## 4. Key Findings

This section presents the main findings from the SQL analysis. The results focus on four aspects: average hourly demand patterns, daily peak-valley gaps, forecast errors, and weekday-weekend demand differences. Together, these findings help explain when electricity demand fluctuates most and how such fluctuations may be connected with potential storage use cases.

### 4.1 Average Hourly Demand Pattern

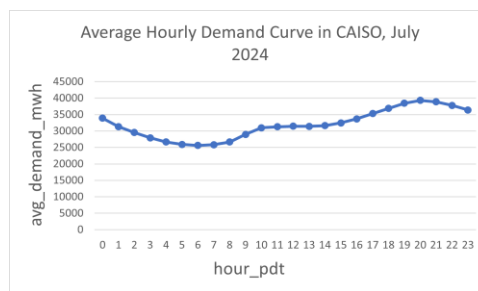


Figure 1. Average Hourly Demand Curve in CAISO, July 2024.

Figure 1 shows the average hourly demand pattern in CAISO during July 2024 after converting the time reference to Pacific Daylight Time (PDT). The demand curve displays clear intraday variation. Average demand is relatively low during the early morning period and rises later in the day. The highest average demand appears during the evening period, suggesting that CAISO faces stronger load pressure during these hours.

This pattern is important for understanding potential storage applications. When demand is low, storage systems may have opportunities to charge. When demand increases during peak hours, storage can discharge to support the grid and reduce peak-load pressure. Therefore, the hourly demand pattern provides a basic indication of possible charging and discharging windows for storage assets.

### 4.2 Daily Peak-Valley Gap

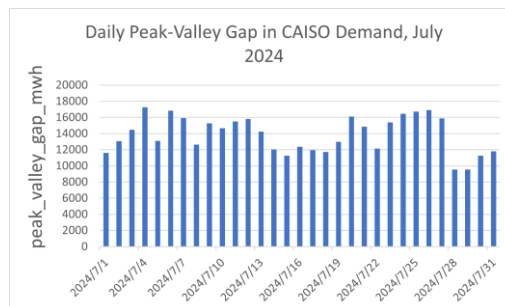


Figure 2. Daily Peak-Valley Gap in CAISO Demand, July 2024.

Figure 2 presents the daily peak-valley gap in CAISO electricity demand during July 2024. The peak-valley gap is calculated as the difference between the highest and lowest demand within each day. A larger gap indicates stronger intraday demand fluctuation and greater pressure on the power system during peak-load periods.

The results show that several days had relatively large peak-valley gaps. For example, July 4 recorded the largest gap, reaching 17,243 MWh, while July 26 and July 6 also showed gaps above 16,800 MWh. These results suggest that demand fluctuation was not evenly distributed across the month. Certain days experienced much stronger intraday variation than others.

From a storage perspective, days with larger peak-valley gaps may provide stronger use cases for battery energy storage. During low-demand hours, storage assets can charge, while during high-demand hours, they can discharge to reduce peak pressure. Therefore, the daily peak-valley analysis provides direct evidence for the potential value of peak shaving and valley filling.

### 4.3 Forecast Error Analysis

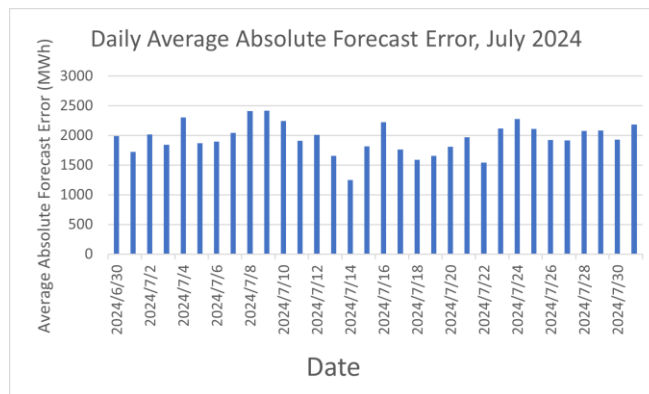


Figure 3. Daily Average Absolute Forecast Error in CAISO Demand, July 2024.

Figure 3 shows the daily average absolute forecast error between actual demand and day-ahead demand forecast. The forecast error was calculated by comparing the actual demand with the day-ahead forecast for each hourly observation. The absolute forecast error was then used to measure the size of the deviation, regardless of whether the forecast was higher or lower than actual demand.

The results indicate that forecast errors varied across the month. Some days showed larger average absolute forecast errors, meaning that actual electricity demand deviated more from the day-ahead forecast on those dates. This suggests that demand uncertainty was not constant throughout July 2024.

Forecast error is important because power system operators need to balance electricity supply and demand in real time. When actual demand differs from forecasted demand, the system may require additional flexible resources to respond to unexpected changes. Battery storage can help provide this flexibility by charging or discharging when actual demand deviates from expectations. Therefore, forecast error analysis adds another perspective to the potential role of storage in grid operation.

#### 4.4 Weekday vs Weekend Demand Pattern

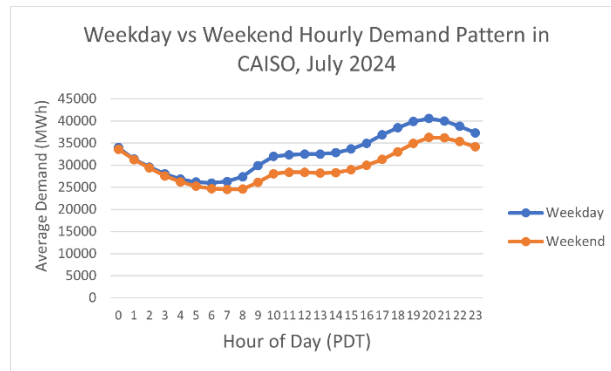


Figure 4. Weekday vs Weekend Hourly Demand Pattern in CAISO, July 2024.

Figure 4 compares the average hourly demand pattern between weekdays and weekends in CAISO during July 2024. The hourly values are shown in Pacific Daylight Time (PDT). Overall, weekday demand was generally higher than weekend demand across most hours of the day, especially during the afternoon and evening periods.

Both weekday and weekend demand curves show clear intraday variation. Demand tends to be lower in the early morning and increases later in the day. The weekday curve reaches a higher evening peak than the weekend curve, suggesting that workday-related economic and social activities may contribute to stronger electricity demand.

This difference is relevant for storage operation. If weekday demand is consistently higher and has a stronger evening peak, storage systems may have greater value during weekday peak-load periods. At the same time, the similar shape of the weekday and weekend curves suggests that storage charging and discharging strategies may still follow a common daily pattern, with charging opportunities during low-demand hours and discharging opportunities during high-demand hours.

#### 5. Storage Use Case Implications

The findings from the demand analysis suggest several potential storage use cases. First, the average hourly demand curve shows that CAISO electricity demand follows a clear intraday pattern, with lower demand during early morning hours and higher demand during evening peak periods. This pattern indicates potential charging and discharging windows for battery storage. Storage systems may charge when demand is relatively low and discharge when demand rises, helping to reduce pressure during peak-load periods.

Second, the daily peak-valley gap analysis shows that some days experienced much larger intraday demand fluctuations than others. These high-gap days may provide stronger opportunities for peak

shaving and valley filling. On such days, storage resources could help smooth demand variation by absorbing electricity during lower-demand hours and providing electricity during peak-demand hours.

Third, the forecast error analysis shows that actual demand does not always match day-ahead forecasts. When actual demand deviates from forecasted demand, grid operators may need flexible resources to respond quickly. Battery storage can provide short-term flexibility by adjusting charging and discharging behavior in response to real-time system conditions.

Finally, the weekday-weekend comparison suggests that storage operation strategies should consider different demand patterns across day types. Weekday demand was generally higher than weekend demand, especially during afternoon and evening hours. This implies that storage assets may have stronger peak-support value during weekdays, while still following a similar daily charging and discharging logic across the week.

Overall, the results do not directly estimate storage revenue or investment returns, but they provide an initial demand-side foundation for understanding where storage may create operational value. Future analysis could combine this demand dataset with electricity price data, renewable generation data, and storage technical parameters to evaluate storage arbitrage, peak shaving value, and grid flexibility benefits more quantitatively.

## **6. Limitations and Future Work**

This project has several limitations. First, the analysis only focuses on July 2024. Since electricity demand can vary significantly across seasons, one month of data is not enough to represent the full annual demand pattern. For example, summer and winter demand may be affected by different weather conditions, cooling needs, and heating needs. A future version of this project could extend the analysis to a full year or multiple years to compare seasonal demand patterns.

Second, this project mainly focuses on demand-side indicators. Although the analysis identifies hourly demand patterns, daily peak-valley gaps, forecast errors, and weekday-weekend differences, it does not include electricity price data. Without price data, the project cannot directly evaluate storage revenue from price arbitrage. Future analysis could combine CAISO demand data with locational marginal price data to estimate potential charging and discharging opportunities more quantitatively.

Third, the project does not include weather variables, renewable generation data, or detailed storage technical parameters. Factors such as temperature, solar generation, wind generation, battery capacity, round-trip efficiency, and charging constraints could all affect storage operation decisions. Adding these variables would make the analysis more realistic and closer to actual power market applications.

Finally, the project is designed as a SQL-based exploratory analysis rather than a predictive or causal model. The results describe demand patterns and potential storage use cases, but they do not establish a causal relationship or estimate investment returns. Future work could use regression analysis or time-

series modeling to study the relationship between demand, weather, renewable generation, and electricity prices.

## 7. Appendix

This appendix summarizes the main project files created during the analysis. The SQL scripts document the workflow from database setup to demand analysis, while the exported CSV files and figures support the key findings discussed in the report.

**Table 2. SQL Script Files**

| SQL File                               | Purpose                                                    |
|----------------------------------------|------------------------------------------------------------|
| 00_database_setup_and_import_check.sql | Create database, create raw table, and check imported data |
| 01_create_demand_view.sql              | Create the demand-only view for July 2024                  |
| 02_daily_peak_valley_analysis.sql      | Calculate daily peak, valley, and peak-valley gap          |
| 03_hourly_demand_pattern_analysis.sql  | Analyze average demand by hour of day                      |
| 04_peak_and_valley_hours_analysis.sql  | Identify peak and valley hours using window functions      |
| 05_forecast_error_analysis.sql         | Compare actual demand with day-ahead demand forecast       |
| 06_weekday_weekend_analysis.sql        | Compare weekday and weekend demand patterns                |
| 07_key_findings_summary.sql            | Summarize key project metrics for reporting                |

**Table 3. Main Output Files**

| Output File                              | Description                                 |
|------------------------------------------|---------------------------------------------|
| fig_01_average_hourly_demand_curve.png   | Average hourly demand curve in PDT          |
| fig_02_daily_peak_valley_gap.png         | Daily peak-valley gap chart                 |
| fig_03_daily_forecast_error.png          | Daily average absolute forecast error chart |
| fig_04_weekday_weekend_hourly_demand.png | Weekday vs weekend hourly demand chart      |
| 04_daily_peak_valley_hours.csv           | Daily peak and valley hour identification   |
| 05_daily_forecast_error_summary.csv      | Daily forecast error summary                |
| 06_hourly_pattern_by_day_type.csv        | Hourly demand pattern by weekday/weekend    |
| 06_weekday_weekend_peak_valley_gap.csv   | Peak-valley gap comparison by day type      |

The full SQL scripts and exported result files are stored in the project folder and can be used to reproduce the analysis.